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United States Patent	12412697
Kind Code	B2
Date of Patent	September 09, 2025
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Coil component

Abstract

A highly reliable coil component having reduced electric resistance and improved bonding strength in a connection portion between a coil conductor and an external electrode. A coil component includes an element body including a coil conductor including a wound conductive wire coated with an insulating film, a magnetic portion containing metal magnetic particles and resin, and an external electrode on a surface of the element body and electrically connected to exposed surfaces, which are exposed on the surface of the element body, of extended portions of the coil conductor. The external electrode includes at least one or more layers. When an average crystal grain size of crystal grains constituting the coil conductor is defined as a and an average crystal grain size of crystal grains constituting a base electrode layer of the external electrode directly connected to the coil conductor is defined as b , $a > b$ is satisfied.

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Appl. No.: 17/929909

Filed: September 06, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220415567 A1	Dec. 29, 2022

Foreign Application Priority Data

JP	2020-053576	Mar. 25, 2020
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Related U.S. Application Data

continuation parent-doc WO PCT/JP2021/012477 20210325 PENDING child-doc US 17929909

Publication Classification

Int. Cl.: **H01F27/29** (20060101); **H01F17/00** (20060101); **H01F17/04** (20060101); **H01F27/28** (20060101)

U.S. Cl.:

CPC **H01F27/292** (20130101); **H01F17/0013** (20130101); **H01F17/04** (20130101); **H01F27/2823** (20130101); H01F2017/048 (20130101)

Field of Classification Search

CPC: H01F (27/292); H01F (17/0013); H01F (17/04); H01F (27/2823); H01F (2017/048)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims benefit of priority to International Patent Application No. PCT/JP2021/012477, filed Mar. 25, 2021, and to Japanese Patent Application No. 2020-053576, filed Mar. 25, 2020, the entire contents of each are incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to a coil component.

Background Art

(3) As an existing coil component, there has been disclosed a coil component including a main body portion having a rectangular parallelepiped shape and a pair of external terminal electrodes provided so as to cover a pair of opposing end surfaces of the main body portion, as described, for example, in Japanese Unexamined Patent Application Publication No. 2016-103598. The main body portion of the coil component includes a coil portion having a substrate and conductor patterns for a planar air-core coil provided on both upper and lower surfaces of the substrate.

SUMMARY

(4) In the coil component disclosed in Japanese Unexamined Patent Application Publication No. 2016-103598, the external terminal electrodes are formed by applying a resin electrode material to the end surfaces and then performing metal plating on the resin electrode material. The pair of external terminal electrodes is connected to the conductor pattern for the planar air-core coil.

(5) Meanwhile, at present, the coil component is required to be miniaturized, and it is important to improve connectivity between a pair of external terminal electrodes and a conductor pattern for a planar air-core coil. When the coil component as disclosed in Japanese Unexamined Patent Application Publication No. 2016-103598 is miniaturized, the contact area between the conductor pattern for the planar air-core coil and the pair of external terminal electrodes is reduced, and thus there is a concern that the connectivity therebetween is reduced. Furthermore, there is a risk that the electric resistance increases, and there is also a problem that the mechanical strength decreases.

(6) Therefore, the present disclosure provides a coil component having high reliability by reducing electric resistance and improving bonding strength at a connection portion between a coil conductor and an external electrode.

(7) A coil component according to the present disclosure is a coil component that includes an element body including a coil conductor formed by winding a conductive wire coated with an insulating film and a magnetic portion containing metal magnetic particles and resin, and an external electrode arranged on a surface of the element body to be electrically connected to an exposed surface of an extended portion of the coil conductor. The exposed surface is exposed on the surface of the element body, in which the external electrode includes at least one or more of layers. When an average crystal grain size of crystal grains constituting the coil conductor is defined as a , and an average crystal grain size of crystal grains constituting the layer of the external electrode directly connected to the coil conductor is defined as b , $a > b$ is satisfied.

(8) In the coil component according to the present disclosure, when the average crystal grain size of the crystal grains constituting the coil conductor is defined as a and the average crystal grain size of the crystal grains constituting the layer of the external electrode directly connected to the coil conductor is defined as b , $a > b$ is satisfied, therefore, an arrangement is easily made such that the shapes of surfaces of the crystal grains of the external electrode follow surfaces of the crystal grains constituting the exposed surface of the coil conductor, whereby inclusion of resin or cavities between the coil conductor and the external electrode is suppressed, electric resistance between the coil conductor and the external electrode can be reduced, and further bonding strength can also be

improved.

(9) According to the present disclosure, it is possible to provide a coil component having high reliability by reducing electric resistance and improving bonding strength at a connection portion between a coil conductor and an external electrode.

(10) The above and other objects, features and advantages of the present disclosure will become more apparent from the following description of embodiments with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an external perspective view schematically illustrating a first embodiment of a coil component of the present disclosure;

(2) FIG. 2 is a transparent perspective view of a magnetic portion in which a coil conductor is embedded in the coil component illustrated in FIG. 1;

(3) FIG. 3 is a cross-sectional view taken along line of FIG. 1, illustrating the coil component according to the present disclosure;

(4) FIG. 4 is a cross-sectional view taken along line IV-IV of FIG. 1, illustrating the coil component according to the present disclosure;

(5) FIG. 5A is a cross-sectional view taken along line V-V of FIG. 1 illustrating the coil component according to the present disclosure, and FIG. 5B is an enlarged cross-sectional view of a portion a;

(6) FIG. 6 is an enlarged cross-sectional view illustrating a first modification of the structure around an extended portion of the coil conductor;

(7) FIG. 7 is an enlarged cross-sectional view illustrating a second modification of the structure around the extended portion of the coil conductor;

(8) FIG. 8 is an enlarged cross-sectional view illustrating a third modification of the structure around the extended portion of the coil conductor;

(9) FIG. 9A is an enlarged cross-sectional view illustrating a fourth modification of the structure around the extended portion of the coil conductor, and FIG. 9B is an enlarged view illustrating the fourth modification of the structure around the extended portion of the coil conductor viewed from an end surface side of an element body, excluding an external electrode;

(10) FIG. 10 is a transparent perspective view illustrating a first modification of the element body of the coil component of the embodiment of the present disclosure;

(11) FIG. 11A is a transparent perspective view illustrating a second modification of the element body of the coil component of the embodiment of the present disclosure, and FIG. 11B is a transparent perspective view when viewed from a direction different from that of FIG. 11A except for a base electrode layer;

(12) FIG. 12A is a cross-sectional view taken along line XIIa-XIIa of FIG. 11B illustrating a coil component according to the present disclosure, and FIG. 12B is an enlarged cross-sectional view of a portion b;

(13) FIG. 13 is an external perspective view schematically illustrating a second embodiment of a coil component of the present disclosure;

(14) FIG. 14 is a transparent perspective view of a magnetic portion in which a coil conductor is embedded in the coil component illustrated in FIG. 13;

(15) FIG. 15A is a cross-sectional view taken along line XVa-XVa of FIG. 14 illustrating the coil component according to the present disclosure, and FIG. 15B is an enlarged cross-sectional view of a portion c;

(16) FIGS. 16A-16D show a manufacturing process diagram illustrating an embodiment of manufacturing a first molded body in a method of manufacturing a coil component; and

(17) FIGS. 17A-17D show a manufacturing process diagram illustrating an embodiment of manufacturing an aggregate substrate in the method of manufacturing the coil component.

DETAILED DESCRIPTION

(18) 1. Coil Component

(19) Hereinafter, a coil component according to a first embodiment of the present disclosure will be described in detail with reference to the drawings.

(20) FIG. 1 is an external perspective view schematically illustrating the first embodiment of the coil component of the present disclosure. FIG. 2 is a transparent perspective view of a magnetic portion in which a coil conductor is embedded in the coil component illustrated in FIG. 1. FIG. 3 is a cross-sectional view taken along line of FIG. 1, illustrating the coil component according to the present disclosure. FIG. 4 is a cross-sectional view taken along line IV-IV of FIG. 1, illustrating the coil component according to the present disclosure. FIG. 5A is a cross-sectional view taken along line V-V of FIG. 1 illustrating the coil component according to the present disclosure, and FIG. 5B is an enlarged cross-sectional view of a portion a.

(21) A coil component **10** includes a rectangular parallelepiped element body **12** and an external electrode **40**.

(22) (A) Element Body

(23) The element body **12** includes a magnetic portion **14** and a coil conductor **16** embedded in the magnetic portion **14**. The element body **12** has a first main surface **12a** and a second main surface **12b** opposite to each other in a pressurization direction x, a first side surface **12c** and a second side surface **12d** opposite to each other in a width direction y orthogonal to the pressurization direction x, and a first end surface **12e** and a second end surface **12f** opposite to each other in a length direction z orthogonal to the pressurization direction x and the width direction y. The dimensions of the element body **12** are not particularly limited.

(24) (B) Magnetic Portion

(25) The magnetic portion **14** contains metal magnetic particles and a resin material.

(26) The resin material is not particularly limited, and examples thereof include thermosetting resins, and organic materials such as epoxy resins, phenol resins, polyester resins, polyimide resins, and polyolefin resins. Only one or two or more types of resin materials may be used.

(27) The metal magnetic particles preferably include first metal magnetic particles and second metal magnetic particles, but may include only the first metal magnetic particles.

(28) The first metal magnetic particles have an average particle size of equal to or more than 10 μm . In addition, the first metal magnetic particles preferably have the average particle size of equal to or less than 200 μm , more preferably equal to or less than 100 μm , and still more preferably equal to or less than 80 μm . By setting the average particle size of the first metal magnetic particles to equal to or more than 10 μm , the magnetic characteristics of the magnetic portion are improved.

(29) The average particle size of the second metal magnetic particles is smaller than that of the first metal magnetic particles. The second metal magnetic particles have the average particle size of equal to or less than 5 μm . As such, when the average particle size of the second metal magnetic particles is smaller than the average particle size of the first metal magnetic particles, the filling property of the metal magnetic particles in the magnetic portion **14** is further improved, and thus the magnetic characteristics of the coil component **10** can be improved.

(30) Here, the average particle size means an average particle size D50 (cumulative percentage of volume basis 50% equivalent particle size). The average particle size D50 can be measured by, for example, a dynamic light scattering particle size analyzer (UPA manufactured by Nikkiso Co., Ltd.).

(31) The first metal magnetic particles and the second metal magnetic particles are not particularly limited, and examples thereof include iron, cobalt, nickel, gadolinium, and an alloy containing one or two or more thereof. Preferably, the first metal magnetic particles and the second metal magnetic particles are iron or an iron alloy. The iron alloy is not particularly limited, and examples thereof

include Fe—Si, Fe—Si—Cr, Fe—Ni, Fe—Si—Al, and the like. The first metal magnetic particles and the second metal magnetic particles of one type or two or more types may be used.

(32) The surfaces of the first metal magnetic particles and the second metal magnetic particles may be covered with an insulating coating. By covering the surfaces of the metal magnetic particles with an insulating coating, the internal resistance of the magnetic portion **14** can be increased. In addition, since the surfaces of the metal magnetic particles are covered by the insulating coating to ensure insulation, it is possible to suppress short-circuit failure with the coil conductor **16**.

(33) Examples of materials for the insulating coating include oxides of silicon, phosphate glass, bismuth glass, and the like. In particular, an insulating coating by zinc phosphate glass in which mechanochemical treatment is performed on the metal magnetic particles is preferable.

(34) The thickness of the insulating coating is not particularly limited, but may be preferably equal to or more than 5 nm and equal to or less than 500 nm (i.e., from 5 nm to 500 nm), more preferably equal to or more than 5 nm and equal to or less than 100 nm (i.e., from 5 nm to 100 nm), and even more preferably equal to or more than 10 nm and equal to or less than 100 nm (i.e., from 10 nm to 100 nm). By further increasing the thickness of the insulating coating, the resistance of the magnetic portion **14** can be further increased. In addition, by further reducing the thickness of the insulating coating, the amount of the metal magnetic particles in the magnetic portion **14** can be further increased, and the magnetic characteristics of the magnetic portion **14** are improved.

(35) The content of the first metal magnetic particles and the second metal magnetic particles in the magnetic portion **14** is preferably equal to or more than 50% by volume, more preferably equal to or more than 60% by volume, and still more preferably equal to or more than 70% by volume with respect to the entire magnetic portion. By setting the content of the first metal magnetic particles and the second metal magnetic particles in such ranges, the magnetic characteristics of the coil component of the present disclosure are improved. In addition, the content of the first metal magnetic particles and the second metal magnetic particles is preferably equal to or less than 99% by volume, more preferably equal to or less than 95% by volume, and still more preferably equal to or less than 90% by volume with respect to the entire magnetic portion **14**. By setting the content of the first metal magnetic particles and the second metal magnetic particles in such ranges, the resistance of the magnetic portion **14** can be further increased.

(36) In the surface portion of the magnetic portion **14**, a region adjacent to the coil conductor **16** may be removed. By removing the magnetic portion **14** in the region adjacent to the coil conductor **16**, a gap between the magnetic portion **14** and the coil conductor **16** is increased, and a medium easily enters during the barrel plating process, so that a plating film is formed over a wider area of the coil conductor **16**. As such, an improvement of bonding strength and a reduction in electric resistance are expected.

(37) (C) Coil Conductor

(38) The coil conductor **16** described above includes a winding portion **20** formed by winding a conductive wire containing a conductive material into a coil shape, a first extended portion **22a** extended to one side of the winding portion **20**, and a second extended portion **22b** extended to the other side of the winding portion **20**.

(39) The winding portion **20** is formed by winding in two stages. The coil conductor **16** is formed by winding a rectangular conductive wire into an α -winding shape. For example, the dimension of the rectangular conductive wire in the width direction y is equal to or more than 15 μm and equal to or less than 200 μm (i.e., from 15 μm to 200 μm), and the dimension thereof in the pressurization direction x is equal to or more than 50 μm and equal to or less than 500 μm (i.e., from 50 μm to 500 μm).

(40) The first extended portion **22a** is exposed from the first end surface **12e** of the element body **12** to form a first exposed portion **24a**, and the second extended portion **22b** is exposed from the second end surface **12f** of the element body **12** to form a second exposed portion **24b**. In the first exposed portion **24a**, an exposed surface of the first extended portion **22a** is formed so as to

intersect an extending direction of the first extended portion **22a**. Further, in the second exposed portion **24b**, the exposed surface of the second extended portion **22b** is formed so as to intersect the extending direction of the second extended portion **22b**.

(41) The coil conductor **16** is formed of a conductive wire such as a metal wire or a wire. The conductive material of the coil conductor **16** is not particularly limited, but is, for example, a metal component made of Ag, Au, Cu, Ni, Sn, or an alloy thereof. Preferably, the conductive material is copper. Only one or two or more types of conductive materials may be used.

(42) As illustrated in FIG. 5B, the coil conductor **16** is made of a plurality of crystal grains **17**. An average crystal grain size a of the crystal grains **17** constituting the coil conductor **16** is preferably more than 2 μm and equal to or less than 10 μm (i.e., from 2 μm to 10 μm).

(43) The average crystal grain size a of the crystal grains **17** constituting the coil conductor **16** is defined as an average value of equivalent circle diameters of the respective crystal grains **17** obtained by observing each coil conductor **16** cross-sectional fabricated by a focused ion beam (FIB) or cross-section ion milling (CP: Cross-section Polisher) at a magnification of 1000 times or more with a transmission electron microscope (TEM: Tunneling Electron Microscope) or a scanning electron microscope (SEM) and selecting 10 or more crystal grains **17**.

(44) The cross section described above is a cross section of the coil conductor **16** in the vicinity of a connection portion with the external electrode **40**. A cross section extending over both of the coil conductor **16** and the external electrode **40** is more preferable.

(45) The vicinity of the surface of the coil conductor **16** is excluded from the measurement target because there is a possibility that the shapes of the crystal grains are deformed.

(46) The surface of the conductive wire constituting the coil conductor **16** is coated with an insulating material to form an insulating film **18**. By coating the conductive wire constituting the coil conductor **16** with an insulating material, it is possible to more reliably insulate the wound coil conductor **16** from each other and insulate the coil conductor **16** from the magnetic portion **14**.

(47) Note that the insulating film **18** is not formed on each of the first exposed portion **24a** and the second exposed portion **24b** of the conductive wire constituting the coil conductor **16**. Therefore, it is easy to form a first base electrode layer **42a** and a second base electrode layer **42b** as a plating layer by plating. In addition, since the coil conductor **16** and the first base electrode layer **42a** and the second base electrode layer **42b** can be connected to each other over wide areas, the electric resistance can be further reduced and the bonding strength can be further improved.

(48) The insulating material of the insulating film **18** is not particularly limited, and examples thereof include polyurethane resin, polyester resin, epoxy resin, polyamide-imide resin, and polyimide resin. Preferably, the insulating film **18** is made of a polyamide-imide resin as an example.

(49) The thickness of the insulating film **18** is preferably equal to or more than 2 μm and equal to or less than 10 μm (i.e., from 2 μm to 10 μm).

(50) As illustrated in FIG. 5B, each of a plurality of concave portions **28** is formed in a surface **26a1** on the first main surface **12a** side and a surface **26a2** on the second main surface **12b** side of the first extended portion **22a** of the conductive wire constituting the coil conductor **16**. Metal magnetic particles **15** and the insulating film **18** are arranged in the concave portions **28**.

Alternatively, only the metal magnetic particles **15** are arranged in the concave portions **28**. At this time, when the metal magnetic particles **15** are arranged in the concave portions **28**, the metal magnetic particles **15** may penetrate the insulating film **18** formed on the surface **26a1** on the first main surface **12a** side and the surface **26a2** on the second main surface **12b** side of the first extended portion **22a**, but preferably do not penetrate the insulating film **18**.

(51) Similarly, each of the plurality of concave portions **28** is formed in a surface **26b1** of the first main surface **12a** and a surface **26b2** on the second main surface side of the second extended portion **22b** of the conductive wire constituting the coil conductor **16**. The metal magnetic particles **15** and the insulating film **18** are arranged in the concave portions **28**. Alternatively, only the metal

magnetic particles **15** are arranged in the concave portions **28**. At this time, when the metal magnetic particles **15** are arranged in the concave portions **28**, the metal magnetic particles **15** may penetrate the insulating film **18** formed on the surface **26b1** on the first main surface **12a** side and the surface **26b2** on the second main surface **12b** side of the second extended portion **22b**, but preferably do not penetrate the insulating film **18**.

(52) In addition, the insulating film **18** is not arranged on the exposed portions (exposed surfaces) of the first exposed portion **24a** and the second exposed portion **24b** of the coil conductor **16** at both the end surfaces **12e** and **12f** of the element body **12**. As such, the coil conductor **16** can be directly electrically connected to the first base electrode layer **42a** and the second base electrode layer **42b**, so that electric resistance between the coil conductor **16** and the first base electrode layer **42a** and the second base electrode layer **42b** can be reduced.

(53) Furthermore, in the metal magnetic particles **15** in contact with the external electrode **40**, the average thickness of the insulating coating in contact with the external electrode **40** is preferably smaller than the average thickness of the insulating coating not in contact with the external electrode **40**. Accordingly, when the external electrode **40** is formed by plating, the metal magnetic particles **15** located around the first extended portion **22a** and the second extended portion **22b** of the coil conductor **16** exposed at the first end surface **12e** and the second end surface **12f** of the element body **12** can be intensively energized to allow plating growth.

(54) Note that the structure around the exposed surface of the extended portion of the coil conductor **16** that is exposed on the surface of the element body **12** may be configured as described below.

(55) FIG. **6** is an enlarged cross-sectional view illustrating a first modification of the structure around the extended portion of the coil conductor **16**.

(56) As illustrated in FIG. **6**, in the first extended portion **22a** and the second extended portion **22b** of the coil conductor **16**, an insulating film removed portion **30** not having the insulating film **18** is formed toward both the end surfaces **12e** and **12f** of the element body **12**. Each of the plurality of concave portions **28** is formed in the surface **26a1** on the first main surface **12a** side and the surface **26a2** on the second main surface **12b** side of the first extended portion **22a** of the coil conductor **16**, and the metal magnetic particles **15** are arranged in the concave portions **28**. Similarly, each of the plurality of concave portions **28** is formed in the surface **26b1** on the first main surface **12a** side and the surface **26b2** on the second main surface **12b** side of the second extended portion **22b** of the coil conductor **16**, and the metal magnetic particles **15** are arranged in the concave portions **28**. Therefore, in the insulating film removed portion **30**, the metal magnetic particles **15** are directly arranged in the concave portions **28** without penetrating the insulating film **18**.

(57) As described above, since the insulating film **18** acts as a cushion when the concave portions **28** are formed in the surface of the coil conductor **16** by the metal magnetic particles **15**, the insulating film **18** acts in a direction in which formation of the concave portions **28** is inhibited, but the concave portions **28** can be easily formed in the surface of the coil conductor **16** by removing the insulating film **18**.

(58) In addition, a part of the external electrode **40** is preferably arranged in the insulating film removed portion **30**. Accordingly, since the coil conductor **16** and the external electrode **40** can be connected to each other over a wide area, it is possible to further reduce the electric resistance and to further improve the bonding strength.

(59) FIG. **7** is an enlarged cross-sectional view illustrating a second modification of the structure around the extended portion of the coil conductor **16**.

(60) In the second modification around the extended portion of the coil conductor **16**, as illustrated in FIG. **7**, in the first extended portion **22a** and the second extended portion **22b** of the coil conductor **16**, the insulating film removed portion **30** not having the insulating film **18** is formed toward both the end surfaces **12e** and **12f** of the element body **12**, similarly to the first modification of the structure around the extended portion of the coil conductor **16**. Further, a minute uneven

portion **32** is formed on surfaces of exposed portions of the first exposed portion **24a** and the second exposed portion **24b** of the coil conductor **16** on both the end surfaces **12e** and **12f** of the element body **12**. As such, since the contact surface area between the coil conductor **16** and the external electrode **40** can be increased, the bonding strength between the coil conductor **16** and the external electrode **40** can be further improved.

(61) FIG. **8** is an enlarged cross-sectional view illustrating a third modification of the structure around the extended portion of the coil conductor **16**.

(62) In the third modification around the extended portion of the coil conductor **16**, as illustrated in FIG. **8**, the insulating film removed portion **30** not having the insulating film **18** is formed in the first extended portion **22a** and the second extended portion **22b** of the coil conductor **16** toward both the end surfaces **12e** and **12f** of the element body **12**, similarly to the first modification of the structure around the extended portion of the coil conductor **16**. Further, a recessed portion **34** is formed in the element body **12** around the exposed portions of the first exposed portion **24a** and the second exposed portion **24b** of the coil conductor **16** on both the end surfaces **12e** and **12f** of the element body **12**. The recessed portion **34** is formed such that an average distance between the coil conductor **16** and the magnetic portion **14** increase toward the direction in which the first extended portion **22a** and the second extended portion **22b** of the coil conductor **16** are extended to both the end surfaces **12e** and **12f**. Accordingly, the external electrode **40** can be arranged so as to fill the recessed portion **34** formed in the peripheral portions of the exposed portions of the first exposed portion **24a** and the second exposed portion **24b** of the coil conductor **16** on both the end surfaces **12e** and **12f** of the element body **12**, therefore, the bonding strength between the coil conductor **16** and the external electrode **40** can be further improved.

(63) FIG. **9** is an enlarged cross-sectional view illustrating a fourth modification of the structure around the extended portion of the coil conductor **16**.

(64) In the fourth modification around the extended portion of the coil conductor **16**, as illustrated in FIG. **9**, the insulating film removed portion **30** not having the insulating film **18** is formed in the first extended portion **22a** and the second extended portion **22b** of the coil conductor **16** toward both the end surfaces **12e** and **12f** of the element body **12**, similarly to the first modification of the structure around the extended portion of the coil conductor **16**. Further, a groove portion **36** having a predetermined width in the width direction *y* is formed at a central portion in the pressurization direction *x* of both the end surfaces **12e** and **12f** of the element body **12** and surfaces (exposed surfaces) of exposed portions of the first exposed portion **24a** and the second exposed portion **24b** of the coil conductor **16** exposed on both the end surfaces **12e** and **12f** of the element body **12**. The depth of the groove portion **36** with respect to the element body **12** is preferably equal to or more than 5 μm and equal to or less than 100 μm (i.e., from 5 μm to 100 μm). Accordingly, the external electrode **40** can be arranged so as to fill the groove portion **36** formed in both the end surfaces **12e** and **12f** of the element body **12** and the surfaces (exposed surfaces) of the exposed portions of the first exposed portion **24a** and the second exposed portion **24b** of the coil conductor **16** exposed on both the end surfaces **12e** and **12f** of the element body **12**, and thus the bonding strength between the coil conductor **16** and the external electrode **40** can be further improved.

(65) (D) External Electrode

(66) The external electrode **40** is arranged on the first end surface **12e** side and the second end surface **12f** side of the element body **12**. The external electrode **40** includes a first external electrode **40a** and a second external electrode **40b**.

(67) The first external electrode **40a** is arranged on the surface of the first end surface **12e** of the element body **12**. Note that the first external electrode **40a** may be formed so as to extend from the first end surface **12e** and cover a part of each of the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d**, or may be extended from the first end surface **12e** to the second main surface **12b** and formed so as to cover a part of each of the first end surface **12e** and the second main surface **12b**. In this case, the first external electrode **40a** is

electrically connected to the first extended portion **22a** of the coil conductor **16**.

(68) The second external electrode **40b** is arranged on the surface of the second end surface **12f** of the element body **12**. Note that the second external electrode **40b** may be formed so as to extend from the second end surface **12f** and cover a part of each of the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d**, or may be extended from the second end surface **12f** to the second main surface **12b** and formed so as to cover a part of each of the second end surface **12f** and the second main surface **12b**. In this case, the second external electrode **40b** is electrically connected to the second extended portion **22b** of the coil conductor **16**.

(69) The thickness of each of the first external electrode **40a** and the second external electrode **40b** is not particularly limited, but may be, for example, equal to or more than 1 μm and equal to or less than 50 μm (i.e., from 1 μm to 50 μm), and preferably equal to or more than 5 μm and equal to or less than 20 μm (i.e., from 5 μm to 20 μm).

(70) The first external electrode **40a** includes the first base electrode layer **42a** and a first plating layer **44a** arranged on the surface of the first base electrode layer **42a**. Similarly, the second external electrode **40b** includes the second base electrode layer **42b** and a second plating layer **44b** arranged on the surface of the second base electrode layer **42b**.

(71) The first base electrode layer **42a** is arranged on the surface of the first end surface **12e** of the element body **12**. Therefore, the first base electrode layer **42a** is in direct contact with the first exposed portion **24a** of the coil conductor **16**. Note that the first base electrode layer **42a** may be formed so as to extend from the first end surface **12e** and cover a part of each of the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d**, or may be formed so as to extend from the first end surface **12e** and cover a part of each of the first end surface **12e** and the second main surface **12b**.

(72) In addition, the second base electrode layer **42b** is arranged on the surface of the second end surface **12f** of the element body **12**. Therefore, the second base electrode layer **42b** is in direct contact with the second exposed portion **24b** of the coil conductor **16**. Note that the second base electrode layer **42b** may be formed so as to extend from the second end surface **12f** and cover a part of each of the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d**, or may be formed so as to extend from the second end surface **12f** and cover a part of each of the second end surface **12f** and the second main surface **12b**.

(73) The first base electrode layer **42a** and the second base electrode layer **42b** are made of a conductive material, preferably one or more metal materials selected from Au, Ag, Pd, Ni, and Cu. The first base electrode layer **42a** and the second base electrode layer **42b** each are formed as a plated electrode. The first base electrode layer **42a** and the second base electrode layer **42b** may be formed by electrolytic plating or electroless plating.

(74) In addition, the main component of the metal material constituting the first base electrode layer **42a** and the second base electrode layer **42b** and the main component of the metal material constituting the coil conductor **16** preferably have the same composition. As such, the metal coupling between the coil conductor **16** and the first base electrode layer **42a** and the second base electrode layer **42b** is further stronger, so that the bonding strength is increased and the DC resistance can be reduced.

(75) The average thicknesses of the first base electrode layer **42a** and the second base electrode layer **42b** are, for example, 10 μm .

(76) The first base electrode layer **42a** and the second base electrode layer **42b** are composed of a plurality of crystal grains **43**. An average crystal grain size b of the crystal grains **43** constituting the first base electrode layer **42a** and the second base electrode layer **42b** is preferably equal to or more than 100 nm and equal to or less than 2000 nm (i.e., from 100 nm to 2000 nm), more preferably equal to or more than 100 nm and equal to or less than 1000 nm (i.e., from 100 nm to 1000 nm), and still more preferably equal to or more than 100 nm and equal to or less than 500 nm.

(i.e., from 100 nm to 500 nm).

(77) The average crystal grain size b of the crystal grains **43** constituting the first base electrode layer **42a** and the second base electrode layer **42b** is defined as an average value of equivalent circle diameters of the respective crystal grains **43** obtained by observing each of the first base electrode layer **42a** and the second base electrode layer **42b** cross-sectional fabricated by a focused ion beam (FIB) or a cross-section ion milling (CP: Cross-section Polisher) at a magnification of 1000 times or more with a tunneling electron microscope (TEM) or a scanning electron microscope (SEM) and selecting 10 or more crystal grains **43**.

(78) The cross section described above is a cross section of the base electrode layers **42a** and **42b** in the vicinity of the connection portion with the coil conductor **16**. A cross section extending over both of the base electrode layers **42a** and **42b** and the coil conductor **16** is more preferable.

(79) The vicinity of the surfaces of the base electrode layers **42a** and **42b** is excluded from the measurement target because there is a possibility that the shapes of the crystal grains are deformed.

(80) The average crystal grain size a of the crystal grains **15** constituting the coil conductor **16** and the average crystal grain size b of the crystal grains **43** constituting the first base electrode layer **42a** and the second base electrode layer **42b** satisfy the relationship of $a > b$. More preferably, the average crystal grain size a of the crystal grains **17** constituting the coil conductor **16** and the average crystal grain size b of the crystal grains **43** constituting the first base electrode layer **42a** and the second base electrode layer **42b** satisfy the relationship of $0.5 \geq b/a$.

(81) The first plating layer **44a** is arranged so as to cover the first base electrode layer **42a**. To be specific, the first plating layer **44a** may be arranged so as to cover the first base electrode layer **42a** arranged on the first end surface **12e**, further extend from the first end surface **12e**, and arranged so as to cover the surface of the first base electrode layer **42a** arranged on the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d**, or may be arranged so as to cover the first base electrode layer **42a** extending from the first end surface **12e** and arranged so as to cover a part of the second main surface **12b**.

(82) The second plating layer **44b** is arranged so as to cover the second base electrode layer **42b**. To be specific, the second plating layer **44b** may be arranged so as to cover the second base electrode layer **42b** arranged on the second end surface **12f**, further extend from the second end surface **12f**, and arranged so as to cover the surface of the second base electrode layer **42b** arranged on the first main surface **12a**, the second main surface **12b**, the first side surface **12c**, and the second side surface **12d**, or may be arranged so as to cover the second base electrode layer **42b** extending from the second end surface **12f** and arranged so as to cover a part of the second main surface **12b**.

(83) The metal material of the first plating layer **44a** and the second plating layer **44b** includes, for example, at least one selected from the group consisting of Cu, Ni, Ag, Sn, Pd, Ag—Pd alloys, Au, and the like.

(84) The first plating layer **44a** and the second plating layer **44b** may be formed in a plurality of layers.

(85) The first plating layer **44a** has a two-layer structure including a first Ni-plating layer **46a** and a first Sn-plating layer **48a** formed on the surface of the first Ni-plating layer **46a**. The second plating layer **44b** has a two-layer structure including a second Ni-plating layer **46b** and a second Sn-plating layer **48b** formed on the surface of the second Ni-plating layer **46b**.

(86) The average thickness of the first Ni-plating layer **46a** and the second Ni-plating layer **46b** is, for example, 5 μm .

(87) In addition, the average thickness of the first Sn-plating layer **48a** and the second Sn-plating layer **48b** is, for example, 10 μm .

(88) Note that the first external electrode **40a** and the second external electrode **40b** may be provided with the following configuration.

(89) For example, the first base electrode layer **42a** and the second base electrode layer **42b** may be an Ag-containing resin electrode, or may be formed of an Ag sputtering layer, a Cu sputtering layer,

or a Ti sputtering layer by sputtering. Note that when the first base electrode layer **42a** and the second base electrode layer **42b** are made of the Ag-containing resin electrode, glass frits may be contained. In addition, when the first base electrode layer **42a** and the second base electrode layer **42b** are formed by sputtering layers, a Cu sputtering layer may be formed on a Ti sputtering layer. (90) In addition, the outermost layers of the first plating layer **44a** and the second plating layer **44b** may be formed only by the Sn-plating layers **48a** and **48b**.

(91) Alternatively, an Ag-plating layer or a Ni-plating layer may be formed on the element body **12** without forming the first base electrode layer **42a** and the second base electrode layer **42b**.

(92) (E) Protection Layer

(93) In this embodiment, a protection layer **50** is provided on the surface of the element body **12** excluding the first exposed portion **24a** exposed on the first end surface **12e** of the element body **12** and the second exposed portion **24b** exposed on the second end surface **12f**. The protection layer **50** is made of a resin material having a high electrical insulation property, such as an acrylic resin, an epoxy resin, a phenol resin, or a polyimide resin. Note that in the present disclosure, the protection layer **50** is provided, but the present disclosure is not limited thereto, and the protection layer **50** is not necessarily be provided.

(94) When the dimension of the coil component **10** in the length direction *z* is defined as dimension *L*, the dimension *L* is preferably equal to or more than 1.0 mm and equal to or less than 12.0 mm (i.e., from 1.0 mm to 12.0 mm). When the dimension of the coil component **10** in the width direction *y* is defined as dimension *W*, the dimension *W* is preferably equal to or more than 0.5 mm and equal to or less than 12.0 mm (i.e., from 0.5 mm to 12.0 mm). When the dimension of the coil component **10** in the pressurization direction *x* is defined as dimension *T*, the dimension *T* is preferably equal to or more than 0.5 mm and equal to or less than 6.0 mm (i.e., from 0.5 mm to 6.0 mm).

(95) According to the coil component **10** illustrated in FIG. 1, since the average crystal grain size *a* of the crystal grains **17** constituting the coil conductor **16** and the average crystal grain size *b* of the crystal grains **43** constituting the first base electrode layer **42a** and the second base electrode layer **42b** in direct contact with the coil conductor **16** satisfy the relationship of $a > b$, the arrangement is easily made such that the crystal grains **43** of the first base electrode layer **42a** follow the shapes of the surfaces of the crystal grains **17** constituting the coil conductor **16** exposed on both the end surfaces **12e** and **12f**, thereby reducing the electric resistance between the coil conductor **16** and the first base electrode layer **42a** and the second base electrode layer **42b**. In addition, the bonding strength between the coil conductor **16** and the first base electrode layer **42a** and the second base electrode layer **42b** can be improved.

(96) In addition, in the coil component **10** illustrated in FIG. 1, the average crystal grain size *a* of the crystal grains **17** constituting the coil conductor **16** and the average crystal grain size *b* of the crystal grains **43** constituting the first base electrode layer **42a** and the second base electrode layer **42b** satisfy the relationship of $0.5 \geq b/a$, whereby the crystal grains **43** of the first base electrode layer **42a** can be arranged so as to follow the shapes of the surfaces of the crystal grains **17** constituting the coil conductor **16** exposed on both the end surfaces **12e** and **12f**, so that the coil conductor **16** and the first base electrode layer **42a** and the second base electrode layer **42b** can be connected without forming voids therebetween, and thus the bonding strength between the coil conductor **16** and the first base electrode layer **42a** and the second base electrode layer **42b** can be further improved and the electric resistance can be reduced.

(97) Next, a first modification of the element body **12** of the coil component **10** of the embodiment of the present disclosure will be described.

(98) FIG. 10 is a transparent perspective view illustrating the first modification of the element body of the coil component of the embodiment of the present disclosure.

(99) As illustrated in FIG. 10, an element body **112** according to the first modification includes a magnetic portion **114** and a coil conductor **116** embedded in the magnetic portion **114**. The element

body **112** has a first main surface **112a** and a second main surface **112b** opposite to each other in the height direction x, a first side surface **112c** and a second side surface **112d** opposite to each other in the width direction y orthogonal to the height direction x, and a first end surface **112e** and a second end surface **112f** opposite to each other in the length direction z orthogonal to the height direction x and the width direction y.

(100) The coil conductor **116** includes a winding portion **120** formed by winding a conductive wire containing a conductive material into a coil shape, a first extended portion **122a** extended to one side of the winding portion **120**, and a second extended portion **122b** extended to the other side of the winding portion **120**.

(101) The first extended portion **122a** is extended and exposed on the first main surface **112a** of the element body **112** to form a first exposed portion **124a**, and the second extended portion **122b** is exposed from the first main surface **112a** of the element body **112** to form a second exposed portion **124b**. In the first exposed portion **124a**, an exposed surface of the first extended portion **122a** is formed so as to intersect the extending direction of the first extended portion **122a**. Further, in the second exposed portion **124b**, the exposed surface of the second extended portion **122b** is formed so as to intersect the extending direction of the second extended portion **122b**.

(102) The coil conductor **116** is formed by a conductive wire similar to that of the coil conductor **16**, and a conductive material similar to that of the coil conductor **16** can be used. In addition, the coil conductor **116** is composed of a plurality of crystal grains. The average crystal grain size a of the crystal grains constituting the coil conductor **116** is preferably greater than 2 μm and equal to or less than 10 μm (i.e., from 2 μm to 10 μm).

(103) As illustrated in FIG. **10**, when the first extended portion **122a** of the coil conductor **116** is exposed from the first main surface **112a**, the first external electrode (not illustrated) is formed so as to cover a part of the first main surface **112a**. In this case, the first external electrode is electrically connected to the first extended portion **122a** of the coil conductor **116**.

(104) In addition, when the second extended portion **122b** of the coil conductor **116** is exposed from the first main surface **112a** as illustrated in FIG. **10**, the second external electrode (not illustrated) is formed so as to cover a part of the first main surface **112a**. In this case, the second external electrode is electrically connected to the second extended portion **122b** of the coil conductor **116**.

(105) The first external electrode includes a first base electrode layer and a first plating layer arranged on a surface of the first base electrode layer. Similarly, the second external electrode includes a second base electrode layer and a second plating layer arranged on a surface of the second base electrode layer.

(106) In the coil conductor **116**, when the first extended portion **122a** of the coil conductor **116** is exposed from the first main surface **112a** as illustrated in FIG. **10**, the first base electrode layer (not illustrated) is formed on a part of the first main surface **112a** so as to cover the first extended portion **122a** of the coil conductor **116**.

(107) In addition, when the second extended portion **122b** of the coil conductor **116** is exposed from the first main surface **112a** as illustrated in FIG. **10**, the second base electrode layer (not illustrated) is formed on a part of the first main surface **112a** so as to cover the second extended portion **122b** of the coil conductor **116**.

(108) In this case, the first base electrode layer and the second base electrode layer are composed of a plurality of crystal grains. The average crystal grain size b of the crystal grains constituting the first base electrode layer and the second base electrode layer is preferably equal to or more than 100 nm and equal to or less than 2000 nm (i.e., from 100 nm to 2000 nm), more preferably equal to or more than 100 nm and equal to or less than 1000 nm (i.e., from 100 nm to 1000 nm), and still more preferably equal to or more than 100 nm and equal to or less than 500 nm (i.e., from 100 nm to 500 nm).

(109) Further, when the first extended portion **122a** of the coil conductor **116** is exposed from the

first main surface **112a** as illustrated in FIG. **10**, the first plating layer (not illustrated) is formed so as to cover the first base electrode layer arranged on the first main surface **112a**.

(110) In addition, when the second extended portion **122b** of the coil conductor **116** is exposed from the first main surface **112a** as illustrated in FIG. **10**, the second plating layer (not illustrated) is formed so as to cover the second base electrode layer arranged on the first main surface **112a**.

(111) Note that as for the structure around the exposed surface of the extended portion of the coil conductor **16** exposed on the surface of the element body **12**, in FIG. **5** to FIG. **9**, the configuration in which the extended portions **22a** and **22b** of the coil conductor **16** are extended and exposed on both the end surfaces **12e** and **12f** has been described, but the present disclosure is not limited thereto, and the structure around the exposed surfaces of the extended portions **122a** and **122b** exposed on the first main surface **112a** side is also the same structure as illustrated in FIG. **10**.

(112) In addition, a second modification of the element body **12** of the coil component **10** of the embodiment of the present disclosure will be described.

(113) FIG. **11A** is a transparent perspective view illustrating the second modification of the element body of the coil component of the embodiment of the present disclosure, and FIG. **11B** is a transparent perspective view when viewed from a direction different from that in FIG. **11A**.

(114) As illustrated in FIGS. **11A** and **11B**, an element body **212** according to the second modification includes a magnetic portion **214** and a coil conductor **216** embedded in the magnetic portion **214**. The magnetic portion **214** includes a first magnetic portion **214a** having a convex portion arranged in the magnetic core portion of the coil conductor **216** and a second magnetic portion **214b** covering the first magnetic portion **214a** and the coil conductor **216**.

(115) The element body **212** is formed in a substantially rectangular parallelepiped shape and has a first main surface **212a** and a second main surface **212b** opposite to each other in the height direction x, a first side surface **212c** and a second side surface **212d** opposite to each other in the width direction y orthogonal to the height direction x, and a first end surface **212e** and a second end surface **212f** opposite to each other in the length direction z orthogonal to the height direction x and the width direction y.

(116) The coil conductor **216** includes a winding portion **220** arranged on one surface side of the first magnetic portion **214a** and formed by winding a conductive wire containing a conductive material into a coil shape, a first extended portion **222a** extended to one side of the winding portion **220**, and a second extended portion **222b** extended to the other side of the winding portion **220**. The first extended portion **222a** is extended to the first end surface **212e** side of the second main surface **212b** of the element body **212**, and at least a part of the surface, among the surfaces of the first extended portion **222a**, parallel to the extending direction of the first extended portion **222a** is exposed from the second main surface **212b** of the element body **212** to form a first exposed portion **224a**. The second extended portion **222b** is extended to the second end surface **212f** side of the second main surface **212b** of the element body **212**, and at least a part of the surface, among the surfaces of the second extended portion **222b**, parallel to the extending direction of the second extended portion **222b** is exposed from the second main surface **212b** of the element body **212** to form a second exposed portion **224b**.

(117) As described above, the first extended portion **222a** may be subjected to forming and arranged on the second main surface **212b** of the element body **212**, and the second extended portion **222b** is subjected to forming and arranged on the second main surface **212b** of the element body **212**. The second main surface **212b** of the element body **212** may be a mounting surface.

(118) A part of the surface of the first extended portion **222a** extended to the second main surface **212b** may be embedded in the element body **212**, and the plurality of concave portions **28** may be formed in the surfaces of the first extended portion **222a** embedded in the element body **212** in the second main surface **212b**. The concave portions **28** are preferably formed in surfaces **26a3** and **26a4** of the first extended portion **222a** arranged perpendicular to the second main surface **212b**, and more preferably formed on the surfaces **26a3** and **26a4** of the first extended portion **222a**

arranged perpendicular to the second main surface **212b** and on the surface **26a2** of the first extended portion **222a** arranged parallel to the extending direction of the first extended portion **222a**. The metal magnetic particles **15** and the insulating film **18** may be arranged in the concave portions **28**. Alternatively, only the metal magnetic particles **15** may be arranged in the concave portions **28**. At this time, when the metal magnetic particles **15** are arranged in the concave portions **28**, the metal magnetic particles **15** may penetrate the insulating film **18** formed on the surfaces **26a2**, **26a3**, and **26a4** of the first extended portion **222a**, but it is preferable that the metal magnetic particles **15** do not penetrate the insulating film **18**.

(119) The insulating film removed portion **30** not having the insulating film **18** may be formed on the surfaces **26a3** and **26a4** of the first extended portion **222a** embedded in the element body **212** in the second main surface **212b**, and the plurality of concave portions **28** may be formed in the surfaces **26a3** and **26a4** of the second extended portion **222b** in the insulating film removed portion **30**. Further, the metal magnetic particles **15** may be arranged in the concave portions **28**.

(120) As described above, when the metal magnetic particles **15** are arranged in the concave portions **28**, peeling of the first extended portion **222a** from the second main surface **212b** is suppressed by the anchor effect.

(121) Similarly, a part of the surface of the second extended portion **222b** extended to the second main surface **212b** may be embedded in the element body **212**, and the plurality of concave portions **28** may be formed in the surfaces of the second extended portion **222b** embedded in the element body **212** in the second main surface **212b**. The concave portions **28** are preferably formed in surfaces **26b3** and **26b4** of the second extended portion **222b** arranged perpendicular to the second main surface **212b**, and more preferably formed in surfaces **26b3** and **26b4** of the first extended portion **222a** arranged perpendicular to the second main surface **212b** and on the surface **26b2** of the second extended portion **222b** arranged parallel to the extending direction of the second extended portion **222b**. The metal magnetic particles **15** and the insulating film **18** may be arranged in the concave portions **28**. Alternatively, only the metal magnetic particles **15** may be arranged in the concave portions **28**. At this time, when the metal magnetic particles **15** are arranged in the concave portions **28**, the metal magnetic particles **15** may penetrate the insulating film **18** formed on the surfaces **26b2**, **26b3**, and **26b4** of the second extended portion **222b**, but it is preferable that the metal magnetic particles **15** do not penetrate the insulating film **18**.

(122) The insulating film removed portion **30** not having the insulating film **18** may be formed on the surfaces **26b3** and **26b4** of the second extended portion **222b** embedded in the element body **212** in the second main surface **212b**, and the plurality of concave portions **28** may be formed in the surfaces **26b3** and **26b4** of the second extended portion **222b** in the insulating film removed portion **30**. Further, the metal magnetic particles **15** may be arranged in the concave portions **28**.

(123) As described above, when the metal magnetic particles **15** are arranged in the concave portions **28**, peeling of the second extended portion **222b** from the second main surface **212b** is suppressed by the anchor effect.

(124) The coil conductor **216** is formed by a conductive wire similar to that of the coil conductor **16**, and a conductive material similar to that of the coil conductor **16** can be used. In addition, the coil conductor **216** is composed of a plurality of crystal grains. The average crystal grain size a of the crystal grains constituting the coil conductor **216** is preferably greater than $2\ \mu\text{m}$ and equal to or less than $10\ \mu\text{m}$ (i.e., from $2\ \mu\text{m}$ to $10\ \mu\text{m}$).

(125) As illustrated in FIG. **11**, when the first extended portion **222a** of the coil conductor **216** is subjected to forming and exposed from the second main surface **212b**, the first external electrode (not illustrated) is formed so as to cover a part of the second main surface **212b**. In this case, the first external electrode is electrically connected to the first extended portion **222a** of the coil conductor **216**.

(126) In addition, as illustrated in FIG. **11**, when the second extended portion **222b** of the coil conductor **216** is subjected to forming and exposed from the second main surface **212b**, the second

external electrode (not illustrated) is formed so as to cover a part of the second main surface **212b**. In this case, the second external electrode is electrically connected to the second extended portion **222b** of the coil conductor **216**.

(127) The first external electrode includes a first base electrode layer and a first plating layer arranged on a surface of the first base electrode layer. Similarly, the second external electrode includes a second base electrode layer and a second plating layer arranged on a surface of the second base electrode layer.

(128) In the coil conductor **216**, when the first extended portion **222a** of the coil conductor **216** is subjected to forming and exposed from the second main surface **212b** as illustrated in FIG. **11**, the first base electrode layer (not illustrated) is formed on a part of the second main surface **212b** so as to cover the first extended portion **222a** of the coil conductor **216**.

(129) In addition, when the second extended portion **222b** of the coil conductor **216** is subjected to forming and exposed from the second main surface **212b** as illustrated in FIG. **11**, the second base electrode layer (not illustrated) is formed on a part of the second main surface **212b** so as to cover the second extended portion **222b** of the coil conductor **216**.

(130) In this case, the first base electrode layer and the second base electrode layer are composed of a plurality of crystal grains. The average crystal grain size b of the crystal grains constituting the first base electrode layer and the second base electrode layer is preferably equal to or more than 100 nm and equal to or less than 2000 nm (i.e., from 100 nm to 2000 nm), more preferably equal to or more than 100 nm and equal to or less than 1000 nm (i.e., from 100 nm to 1000 nm), and still more preferably equal to or more than 100 nm and equal to or more than 500 nm (i.e., from 100 nm to 500 nm).

(131) Further, as illustrated in FIG. **11**, when the first extended portion **222a** of the coil conductor **216** is subjected to forming and directly extended to the second main surface **212b**, although not illustrated, the first extended portion **222a** may be formed so as to cover the first base electrode layer arranged on the second main surface **212b**.

(132) In addition, as illustrated in FIG. **11**, when the second extended portion **222b** of the coil conductor **216** is subjected to forming and directly extended to the second main surface **212b**, although not illustrated, the second extended portion **222b** may be formed so as to cover the second base electrode layer arranged on the second main surface **212b**.

(133) Note that as for the structure around the exposed surface of the extended portion of the coil conductor **16** exposed on the surface of the element body **12**, in FIG. **5** to FIG. **9**, the configuration in which the extended portions **22a** and **22b** of the coil conductor **16** are extended and exposed on both the end surfaces **12e** and **12f** has been described, but the present disclosure is not limited thereto, and the structure around the exposed surfaces of the extended portions **222a** and **222b** exposed on the second main surface **212b** side is also the same structure as illustrated in FIG. **11**.

(134) Next, a coil component **510** of a second embodiment of the present disclosure will be described.

(135) FIG. **13** is an external perspective view schematically illustrating the second embodiment of the coil component of the present disclosure. FIG. **14** is a transparent perspective view of a magnetic portion in which a coil conductor is embedded in the coil component illustrated in FIG. **13**. FIG. **15A** is a cross-sectional view taken along line XVa-XVa of FIG. **13** illustrating the coil component according to the present disclosure, and FIG. **15B** is an enlarged cross-sectional view of a portion c.

(136) As illustrated in FIG. **14**, the element body **512** includes a magnetic portion **514** and a coil conductor **516** embedded in the magnetic portion **514**. The element body **512** has a first main surface **512a** and a second main surface **512b** opposite to each other in the height direction x , a first side surface **512c** and a second side surface **512d** opposite to each other in the width direction y orthogonal to the height direction x , and a first end surface **512e** and a second end surface **512f** opposite to each other in the length direction z orthogonal to the height direction x and the width

direction y.

(137) The coil conductor **516** includes a winding portion **520** formed by winding a conductive wire containing a conductive material into a coil shape, a first extended portion **522a** extended to one side of the winding portion **520**, and a second extended portion **522b** extended to the other side of the winding portion **520**.

(138) The first extended portion **522a** is exposed from the first end surface **512e** of the element body **512** to form a first exposed portion **524a**, and the second extended portion **522b** is exposed from the second end surface **512f** of the element body **512** to form a second exposed portion **524b**. In the first exposed portion **524a**, the exposed surface of the first extended portion **522a** is arranged parallel to the extending direction of the first extended portion **522a**. Further, in the second exposed portion **524b**, the exposed surface of the second extended portion **522b** is arranged parallel to the extending direction of the second extended portion **522b**.

(139) The coil conductor **516** is formed by a conductive wire similar to that of the coil conductor **16**, and a conductive material similar to that of the coil conductor **16** can be used. In addition, the coil conductor **516** is composed of a plurality of crystal grains. The average crystal grain size a of the crystal grains constituting the coil conductor **516** is preferably greater than $2\text{ }\mu\text{m}$ and equal to or less than $10\text{ }\mu\text{m}$ (i.e., from $2\text{ }\mu\text{m}$ to $10\text{ }\mu\text{m}$).

(140) As illustrated in FIG. **14**, when the first extended portion **522a** of the coil conductor **516** is exposed from the first end surface **512e**, the first external electrode **40a** is arranged on the surface of the first end surface **512e** of the element body **512**. Note that the first external electrode **40a** may be formed so as to extend from the first end surface **512e** and cover a part of each of the first main surface **512a**, the second main surface **512b**, the first side surface **512c**, and the second side surface **512d**, or may be extended from the first end surface **512e** to the second main surface **512b** and formed so as to cover a part of each of the first end surface **512e** and the second main surface **512b**. In this case, a first external electrode **40a** is electrically connected to the first extended portion **522a** of the coil conductor **516**.

(141) In addition, as illustrated in FIG. **14**, when the second extended portion **522b** of the coil conductor **516** is exposed from the second end surface **512f**, the second external electrode **40b** is arranged on the surface of the second end surface **512f** of the element body **512**. Note that the second external electrode **40b** may be formed so as to extend from the second end surface **512f** and cover a part of each of the first main surface **512a**, the second main surface **512b**, the first side surface **512c**, and the second side surface **512d**, or may be extended from the second end surface **512f** to the second main surface **512b** and formed so as to cover a part of each of the second end surface **512f** and the second main surface **512b**. In this case, the second external electrode **40b** is electrically connected to the second extended portions **522b** of the coil conductor **516**.

(142) As illustrated in FIG. **15B**, each of the plurality of concave portions **28** is formed in the surface **26a1** on the first main surface **512a** side and the surfaces **26a2** on the second main surface **512b** side of the first extended portion **522a** of the conductive wire constituting the coil conductor **516**. In addition, it is preferable that the concave portions **28** are formed in the surface **26a3** of the first extended portion **522a** arranged perpendicularly to the second main surface **512b**. The metal magnetic particles **15** and the insulating film **18** are arranged in the concave portions **28**.

Alternatively, only the metal magnetic particles **15** are arranged in the concave portions **28**. At this time, when the metal magnetic particles **15** are arranged in the concave portions **28**, the metal magnetic particles **15** may penetrate the insulating film **18** formed on the surface **26a1** on the first main surface **512a** side and the surface **26a2** on the second main surface **512b** side of the first extended portion **522a**, and the surface **26a3** on the second end surface **512f** side of the first extended portion **522a**, but preferably do not penetrate the insulating film **18**.

(143) The insulating film removed portion **30** not having the insulating film **18** may be formed on the surfaces **26a1** and **26a2** of the first extended portion **522a** embedded in the element body **512** in the first end surface **512e**, and the plurality of concave portions **28** may be formed in the surfaces

26a1 and **26a2** of the first extended portion **522a** in the insulating film removed portion **30**.

Further, the metal magnetic particles **15** may be arranged in the concave portions **28**.

(144) As described above, when the metal magnetic particles **15** are arranged in the concave portions **28**, peeling of the first extended portion **522a** from the first end surface **512e** is suppressed by an anchor effect.

(145) Similarly, each of the plurality of concave portions **28** is formed in the surface **26b1** on the first main surface **512a** side and the surface **26b2** on the second main surface **512b** side of the second extended portion **522b** of the conductive wire constituting the coil conductor **516**. In addition, it is preferable that the concave portions **28** are formed in the surface **26b3** of the second extended portion **522b** arranged perpendicularly to the second main surface **512b**. The metal magnetic particles **15** and the insulating film **18** are arranged in the concave portions **28**.

Alternatively, only the metal magnetic particles **15** are arranged in the concave portions **28**. At this time, when the metal magnetic particles **15** are arranged in the concave portions **28**, the metal magnetic particles **15** may penetrate the insulating film **18** formed on the surface **26b1** on the first main surface **512a** side and the surface **26b2** on the second main surface **512b** side of the second extended portions **522b**, and the surface **26b3** on the first end surface **512e** side of the second extended portion **522b**, but preferably do not penetrate the insulating film **18**.

(146) The insulating film removed portion **30** not having the insulating film **18** may be formed on the surfaces **26b1** and **26b2** of the second extended portion **522b** embedded in the element body **512** in the second end surface **512f**, and the plurality of concave portions **28** may be formed in the surfaces **26b1** and **26b2** of the second extended portion **522b** in the insulating film removed portion **30**. Further, the metal magnetic particles **15** may be arranged in the concave portions **28**.

(147) As described above, when the metal magnetic particles **15** are arranged in the concave portions **28**, peeling of the second extended portion **522b** from the second end surface **512f** is suppressed by the anchor effect.

(148) The first external electrode **40a** includes the first base electrode layer **42a** and the first plating layer **44a** arranged on the surface of the first base electrode layer **42a**. Similarly, the second external electrode **40b** includes the second base electrode layer **42b** and the second plating layer **44b** arranged on the surface of the second base electrode layer **42b**.

(149) As illustrated in FIG. 15, when the first extended portion **522a** of the coil conductor **516** is exposed from the first end surface **512e**, the first base electrode layer **42a** is arranged on the surface of the first end surface **512e** of the element body **512**. Therefore, the first base electrode layer **42a** is in direct contact with the first exposed portion **524a** of the coil conductor **516**. Note that the first base electrode layer **42a** may be formed so as to extend from the first end surface **512e** and cover a part of each of the first main surface **512a**, the second main surface **512b**, the first side surface **512c**, and the second side surface **512d**, or may extend from the first end surface **512e** and be formed so as to cover a part of each of the first end surface **512e** and the second main surface **512b**.

(150) In addition, when the second extended portion **522b** of the coil conductor **516** is exposed from the second end surface **512f** as illustrated in FIG. 15, the second base electrode layer **42b** is arranged on the surface of the second end surface **512f** of the element body **512**. Therefore, the second base electrode layer **42b** is in direct contact with the second exposed portion **524b** of the coil conductor **516**. Note that the second base electrode layer **42b** may be formed so as to extend from the second end surface **512f** and cover a part of each of the first main surface **512a**, the second main surface **512b**, the first side surface **512c**, and the second side surface **512d**, or may extend from the second end surface **512f** and be formed so as to cover a part of each of the second end surface **512f** and the second main surface **512b**.

(151) In this case, the first base electrode layer **42a** and the second base electrode layer **42b** are composed of a plurality of crystal grains. The average crystal grain size b of the crystal grains constituting the first base electrode layer **42a** and the second base electrode layer **42b** is preferably equal to or more than 100 nm and equal to or less than 2000 nm (i.e., from 100 nm to 2000 nm),

more preferably equal to or more than 100 nm and equal to or less than 1000 nm (i.e., from 100 nm to 1000 nm), and still more preferably equal to or more than 100 nm and equal to or less than 500 nm (i.e., from 100 nm to 500 nm).

(152) Further, as illustrated in FIG. 15, when the first extended portion **522a** of the coil conductor **516** is exposed from the first end surface **512e**, the first plating layer **44a** is arranged so as to cover the first base electrode layer **42a**. To be specific, the first plating layer **44a** may be arranged so as to cover the first base electrode layer **42a** arranged on the first end surface **512e**, further extend from the first end surface **512e**, and arranged so as to cover the surface of the first base electrode layer **42a** arranged on the first main surface **512a**, the second main surface **512b**, the first side surface **512c**, and the second side surface **512d**, or may extend from the first end surface **512e**, and be arranged so as to cover the first base electrode layer **42a** arranged so as to cover a part of each of the first end surface **512e** and the second main surface **512b**.

(153) In addition, as illustrated in FIG. 15, when the second extended portion **522b** of the coil conductor **516** is exposed from the second end surface **512f**, the second plating layer **44b** is arranged so as to cover the second base electrode layer **42b**. To be specific, the second plating layer **44b** may be arranged so as to cover the second base electrode layer **42b** arranged on the second end surface **512f**, further extend from the second end surface **512f**, and arranged so as to cover the surface of the second base electrode layer **42b** arranged on the first main surface **512a**, the second main surface **512b**, the first side surface **512c**, and the second side surface **512d**, or may extend from the second end surface **512f**, and be arranged so as to cover the second base electrode layer **42b** arranged so as to cover a part of each of the second end surface **512f** and the second main surface **512b**.

(154) In addition, the minute uneven portion **32** is formed on surfaces of portions in which the first exposed portion **524a** and the second exposed portion **524b** of the coil conductor **516** are exposed on both the end surfaces **512e** and **512f** of the element body **512**. Accordingly, since the contact surface area between the coil conductor **516** and the external electrode **40** can be increased, the bonding strength between the coil conductor **516** and the external electrode **40** can be further improved. The uneven portion **32** can be formed by, for example, irradiating the first exposed portion **524a** and the second exposed portion **524b** of the coil conductor **516** with a laser to form a plurality of holes.

(155) Note that as for the structure around the exposed surface of the extended portion of the coil conductor **16** exposed on the surface of the element body **12**, in FIG. 5 to FIG. 9, the configuration in which the extended portions **22a** and **22b** of the coil conductor **16** are extended and exposed on both the end surfaces **12e** and **12f** has been described, but the present disclosure is not limited thereto, and the structure around the exposed surfaces of the extended portions **522a** and **522b** exposed on both the end surfaces **512e** and **512f** has also the same structure as illustrated in FIG. 14.

(156) 2. Method of Manufacturing Coil Component

(157) Next, a method of manufacturing the coil component will be described.

(158) (A) Preparation of Metal Magnetic Particles

(159) First, metal magnetic particles are prepared. Here, the metal magnetic particles are not particularly limited, and for example, Fe-based soft magnetic material powders such as α -Fe, Fe—Si, Fe—Si—Cr, Fe—Si—Al, Fe—Ni, and Fe—Co can be used. Also, the material form of the metal magnetic particles is preferably amorphous having good soft magnetic properties, but is not particularly limited, and may be crystalline.

(160) The average particle size of the metal magnetic particles is also not particularly limited, but it is preferable to use two or more kinds of metal magnetic particles having different average particle sizes. That is, the metal magnetic particles are dispersed in the resin material. Therefore, from the viewpoint of improving the filling efficiency of the metal magnetic particles, it is preferable to use metal magnetic particles having different average particle sizes, for example, the first metal

magnetic particles having the average particle size of equal to or more than 10 μm and equal to or less than 40 μm (i.e., from 10 μm to 40 μm), and the second metal magnetic particles having the average particle size of equal to or more than 1 μm and equal to or less than 20 μm (i.e., from 1 μm to 20 μm).

(161) (B) Formation of Insulating Coating

(162) Next, the surfaces of the metal magnetic particles are coated with an insulating coating. Here, when the insulating coating is formed by a mechanical method, the metal magnetic particles and the insulating material powder are fed into a rotary vessel, and composite of particles is performed by a mechanochemical treatment, whereby the insulating coating can be formed on the surface of the magnetic powder.

(163) (C) Production of Magnetic Sheet

(164) Next, a resin material is prepared. The resin material is not particularly limited, and for example, epoxy resin, phenol resin, polyester resin, polyimide resin, polyolefin resin, or the like can be used.

(165) Subsequently, the metal magnetic particles coated with the insulating coating and other filler components (glass material, ceramic powder, ferrite powder, and the like) are mixed with a resin material to form a slurry, and then the slurry is subjected to molding processing using a doctor blade method or the like, followed by drying, thereby producing a magnetic sheet having a thickness of equal to or more than 50 μm and equal to or less than 300 μm (i.e., from 50 μm to 300 μm) in which the filler components are dispersed in the resin material.

(166) (D) Production of Aggregate Substrate

(167) Next, Cu is used as a wire conductor, and the coil conductor **16** having an α -winding shape formed of a rectangular conductive wire coated with the insulating film **18** is prepared. When necessary, the insulating film **18** in a region of 50 μm from the end of the coil conductor **16** is removed with a nipper-shaped clamp. Accordingly, the insulating film removed portion **30**, which is a portion not covered with the insulating film **18**, is formed in an annular shape with the extending direction of the coil conductor **16** as the central axis. Note that the insulating film **18** can be removed by burning off by heating, or may be dissolved by chemical or laser. At this time, the concave portions **28** may be provided in advance in the first extended portion **22a** and the second extended portion **22b** of the coil conductor **16**.

(168) Subsequently, the element body **12** in which the coil conductor **16** is embedded is manufactured.

(169) FIGS. **16A-16D** show a manufacturing process diagram illustrating an embodiment of manufacturing a first molded body in a method of manufacturing a coil component. FIGS. **17A-17D** show a manufacturing process diagram illustrating an embodiment of manufacturing an aggregate substrate in the method of manufacturing the coil component.

(170) First, as illustrated in FIG. **16A**, a first mold **60** is prepared, and the coil conductor **16** is arranged on the first mold **60** in a matrix.

(171) Next, as illustrated in FIG. **16B**, a first magnetic sheet **70a** of a mixture containing the first metal magnetic particles, the second metal magnetic particles, and a resin material is laminated on the coil conductor **16**, and then, as illustrated in FIG. **16C**, a second mold **62** is arranged on the upper surface side of the first magnetic sheet **70a**. Then, as illustrated in FIG. **16D**, the first magnetic sheet **70a** is sandwiched between the coil conductor **16** on the first mold **60** and the second mold **62** to perform a primary press molding. By this primary press molding, at least a portion of the coil conductor **16** is embedded in the above-described sheet, and the mixture fills the inside of the coil conductor **16**, whereby a first molded body **72** is produced.

(172) Next, as illustrated in FIG. **17A**, the first molded body **72** obtained by the primary press molding, in which the coil conductor **16** is embedded, is detached from the second mold **62**, the first molded body **72** is turned over, and the first molded body **72** is arranged on the first mold **60**. Then, another second magnetic sheet **70b** is laminated on the surface where the coil conductor **16** is

exposed. Next, as illustrated in FIG. 17B, a third mold **64** is arranged on the upper surface side of the second magnetic sheet **70b**. Then, as illustrated in FIG. 17C, the second magnetic sheet **70b** is sandwiched between the first molded body **72** on the first mold **60** and the third mold **64**, and secondary pressing is performed.

(173) Note that in the third mold **64**, convex portions **64a** and **64b** may be arranged at portions corresponding to the respective extended portions. When the convex portions **64a** and **64b** are arranged, the convex portions **64a** and **64b** can apply more pressure on the vicinity of the extended portion via the second magnetic sheet at the time of the secondary pressing. Therefore, as such, in the secondary pressing illustrated in FIG. 17C, the metal magnetic particles **15** and the insulating film **18** can be sunk in the surface of the first extended portion **22a** and the surface of the second extended portion **22b** of the coil conductor **16**.

(174) Note that alternatively, in the secondary pressing, the metal magnetic particles can be arranged so as to be sunk in by adjusting the pressure at the time of pressurization or by providing concave portions in advance in the surface of the extended portion.

(175) Subsequently, after the secondary pressing, as illustrated in FIG. 17D, the third mold **64** is detached, and an aggregate substrate (second molded body) **74** in which the entire coil conductor **16** is embedded in the first magnetic sheet **70a** and the second magnetic sheet **70b** is produced.

(176) (E) Production of Element Body

(177) Next, the first mold **60** and the third mold **64** are removed, and as illustrated in FIG. 17D, after an aggregate substrate **74** is produced, the aggregate substrate **74** is cut along cut lines by using a cutter such as a dicer to be divided into individual pieces, thereby producing the element body **12** in which the coil conductor **16** is embedded so that the first exposed portion **24a** and the second exposed portion **24b** of the coil conductor **16** are exposed from both end surfaces of the element body **12**. The aggregate substrate **74** can be divided into the element bodies **12** using a dicing blade, various laser devices, a dicer, various cutting tools, or a mold. In a preferred aspect, the cut surface of each element body **12** is barrel-polished.

(178) Next, the protection layer **50** is formed on the entire surface of the element body obtained as described above. The protection layer **50** can be formed by electrodeposition coating, a spray method, a dip method, or the like.

(179) Peripheries of areas where the first exposed portion **24a** and the second exposed portion **24b** in the coil conductor **16** of the element body **12** coated with the protection layer **50** obtained as described above are arranged are irradiated with a laser beam, thereby removing the insulating film **18**, the metal magnetic particles **15** and the insulating coating that coats the metal magnetic particles **15**, and the protection layer **50** around the areas where the first exposed portion **24a** and the second exposed portion **24b** in the coil conductor **16** are arranged, and melting the metal magnetic particles **15**. Note that alternatively, the protection layer **50** can also be removed by melting with a chemical, heating, peeling, a blast treatment, polishing, or the like, other than laser irradiation.

(180) (F) Formation of External Electrodes

(181) Next, the first external electrode **40a** is formed on the first end surface **12e** of the element body **12**, and the second external electrode **40b** is formed on the second end surface **12f**.

(182) First, Cu plating is performed on the element body **12** by electrolytic barrel plating, and a base electrode layer as a plating layer is formed. Subsequently, a Ni-plating layer is formed on the surface of the base electrode layer by Ni plating, further an Sn-plating layer is formed by Sn plating, and the external electrode **40** is formed. As such, the first exposed portion **24a** of the coil conductor **16** is electrically connected to the first external electrode **40a**, and the second exposed portion **24b** of the coil conductor **16** is electrically connected to the second external electrode **40b**. Note that the base electrode layer made by Cu plating may be formed by electroless plating. By forming the base electrode layer by electrolytic plating or electroless plating, the average crystal grain size b of the crystal grains **43** constituting the base electrode layer can be made smaller than

the average crystal grain size **a** of the crystal grains **17** constituting the coil conductor **16**. When formed by electrolytic barrel plating, the size of the crystal grains **43** constituting the base electrode layer can be reduced to obtain a desired average crystal grain size **b** by reducing the size of the medium or reducing the current value to reduce the current density. Alternatively, when the base electrode layer is formed by electrolytic plating or electroless plating, the size of the crystal grains **43** may be reduced by containing an additive such as organic carboxylic acids or organic sulfonic acids in order to obtain a desired average crystal grain size **b**.

(183) As described above, the coil component **10** is manufactured.

(184) Note that the metal magnetic particles **15** of the magnetic portion **14** may be arranged in the concave portion formed on the surface of the conductive wire in the winding portion **20** of the coil conductor **16** inside the element body **12**.

(185) The metal magnetic particles **15** of the magnetic portion **14** are arranged in the concave portions **28** formed in the surface of the first exposed portion **24a** and the surface of the second exposed portion **24b** of the coil conductor **16** from which the insulating film **18** is removed, and thus an anchor effect is generated by the metal magnetic particles **15**, whereby the bonding strength between the magnetic portion **14** and the coil conductor **16** can be improved.

(186) In addition, since the coil conductor **16** and the external electrode **40** are directly bonded to each other, the electric resistance can be reduced.

(187) Note that as described above, the embodiments of the present disclosure are disclosed in the above description, but the present disclosure is not limited thereto.

(188) That is, without departing from the scope of the technical idea and object of the present disclosure, various modifications can be made to the above-described embodiments in terms of mechanism, shape, material, quantity, position, arrangement, and the like, and these are included in the present disclosure.

Claims

1. A coil component comprising: an element body including a coil conductor including a wound conductive wire coated with an insulating film, and a magnetic portion containing metal magnetic particles and resin; and an external electrode on a surface of the element body and electrically connected to an exposed surface of an extended portion of the coil conductor, the exposed surface being exposed on the surface of the element body, wherein the external electrode includes at least one or more layers, and when an average crystal grain size of crystal grains constituting the coil conductor is defined as **a**, and an average crystal grain size of crystal grains constituting the layer of the external electrode directly connected to the coil conductor is defined as **b**, $a > b$ is satisfied.
2. The coil component according to claim 1, wherein the layer of the external electrode directly connected to the coil conductor is a plating layer.
3. The coil component according to claim 2, wherein a metal component constituting the coil conductor and a metal component constituting the layer of the external electrode connected to the coil conductor are metal components having the same composition of main components.
4. The coil component according to claim 2, wherein $0.5 \geq b/a$ is satisfied.
5. The coil component according to claim 2, wherein an insulating film removed portion that is absent of the insulating film is at an extended portion of the coil conductor, and the insulating film removed portion and the external electrode are directly connected to each other.
6. The coil component according to claim 2, wherein an uneven portion is on a surface of the exposed surface.
7. The coil component according to claim 2, wherein a recessed portion is in the element body around the exposed surface such that an average distance between the coil conductor and the magnetic portion increases in a direction in which the coil conductor is extended to a surface of the element body.

8. The coil component according to claim 2, wherein a groove portion having a depth of equal to from 5 μm to 100 μm is in the exposed surface.
 9. The coil component according to claim 2, wherein an extended portion of the coil conductor is extended to a mounting surface of the element body, a plurality of concave portions is in a surface arranged perpendicular to a mounting surface of the element body among surfaces of an extended portion of the coil conductor, and the metal magnetic particles are in the concave portions.
 10. The coil component according to claim 2, wherein an extended portion of the coil conductor is extended to a mounting surface of the element body, and an extending direction of an extended portion of the coil conductor is parallel to a mounting surface of the element body.
 11. The coil component according to claim 1, wherein a metal component constituting the coil conductor and a metal component constituting the layer of the external electrode connected to the coil conductor are metal components having the same composition of main components.
 12. The coil component according to claim 11, wherein $0.5 \geq b/a$ is satisfied.
 13. The coil component according to claim 11, wherein an insulating film removed portion that is absent of the insulating film is at an extended portion of the coil conductor, and the insulating film removed portion and the external electrode are directly connected to each other.
 14. The coil component according to claim 1, wherein $0.5 \geq b/a$ is satisfied.
 15. The coil component according to claim 1, wherein an insulating film removed portion that is absent of the insulating film is at an extended portion of the coil conductor, and the insulating film removed portion and the external electrode are directly connected to each other.
 16. The coil component according to claim 1, wherein an uneven portion is on a surface of the exposed surface.
 17. The coil component according to claim 1, wherein a recessed portion is in the element body around the exposed surface such that an average distance between the coil conductor and the magnetic portion increases in a direction in which the coil conductor is extended to a surface of the element body.
 18. The coil component according to claim 1, wherein a groove portion having a depth of equal to from 5 μm to 100 μm is in the exposed surface.
 19. The coil component according to claim 1, wherein an extended portion of the coil conductor is extended to a mounting surface of the element body, a plurality of concave portions is in a surface arranged perpendicular to a mounting surface of the element body among surfaces of an extended portion of the coil conductor, and the metal magnetic particles are in the concave portions.
 20. The coil component according to claim 1, wherein an extended portion of the coil conductor is extended to a mounting surface of the element body, and an extending direction of an extended portion of the coil conductor is parallel to a mounting surface of the element body.
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